Methodology Summary — Track 3: Semiconductor Wafer Quality & Fault Detection

1. Objective and dataset

The objective is binary classification of semiconductor wafers: target=0 indicates pass and target=1 indicates a defective wafer. The uploaded advanced-track dataset contains 1,253 observations and 590 sensor features. It contains 1,170 passing wafers and 83 defective wafers, a defect rate of 6.62%. Sensor data contains 33,940 missing cells (4.59%).

Schema note: the problem statement lists sensor_0 through sensor_590 (591 sensors), but the uploaded advanced dataset contains sensor_0 through sensor_589 (590 sensors). The inference script aligns incoming data to the actual training schema.

2. Data preparation and model

The timestamp is retained for traceability and excluded from the baseline classifier. Sensor values are converted to numeric values; invalid values become missing. Median imputation and zero-variance feature filtering are placed inside a scikit-learn pipeline so preprocessing is learned from training data. An ExtraTrees ensemble with class_weight='balanced' handles nonlinear patterns while accounting for class imbalance.

3. Validation and results

A stratified 75/25 train/validation split was created with random_state=42. The classification threshold was selected using 5-fold stratified out-of-fold predictions on the training partition; the holdout was not used to select it. PR-AUC is Average Precision computed from predicted probabilities.

Metric	Local validation result
PR-AUC (Average Precision)	0.1973
Balanced Accuracy	0.6164
Defective-class precision	0.0936
Defective-class recall	0.7619
Defective-class F1	0.1667
OOF-selected threshold	0.052
Confusion matrix (TN, FP, FN, TP)	[138, 155, 5, 16]

These are local validation estimates, not official hidden-test results. Precision is low at the selected threshold, indicating false alarms remain a limitation. Threshold selection trades precision against recall; the right operating point depends on the cost of missed defects versus extra inspections.

4. Practical implications, interpretability and limitations

The model could flag wafers for additional inspection before downstream packaging. Feature importance from the fitted ensemble can help prioritize sensor investigation, but importance is associative rather than causal and should be validated by process engineers. The dataset is small relative to its sensor count and highly imbalanced; a single holdout score can vary. Repeated stratified validation, comparison with other classifiers, missingness analysis, and probability calibration are sensible next steps. Hidden-test performance is unknown.